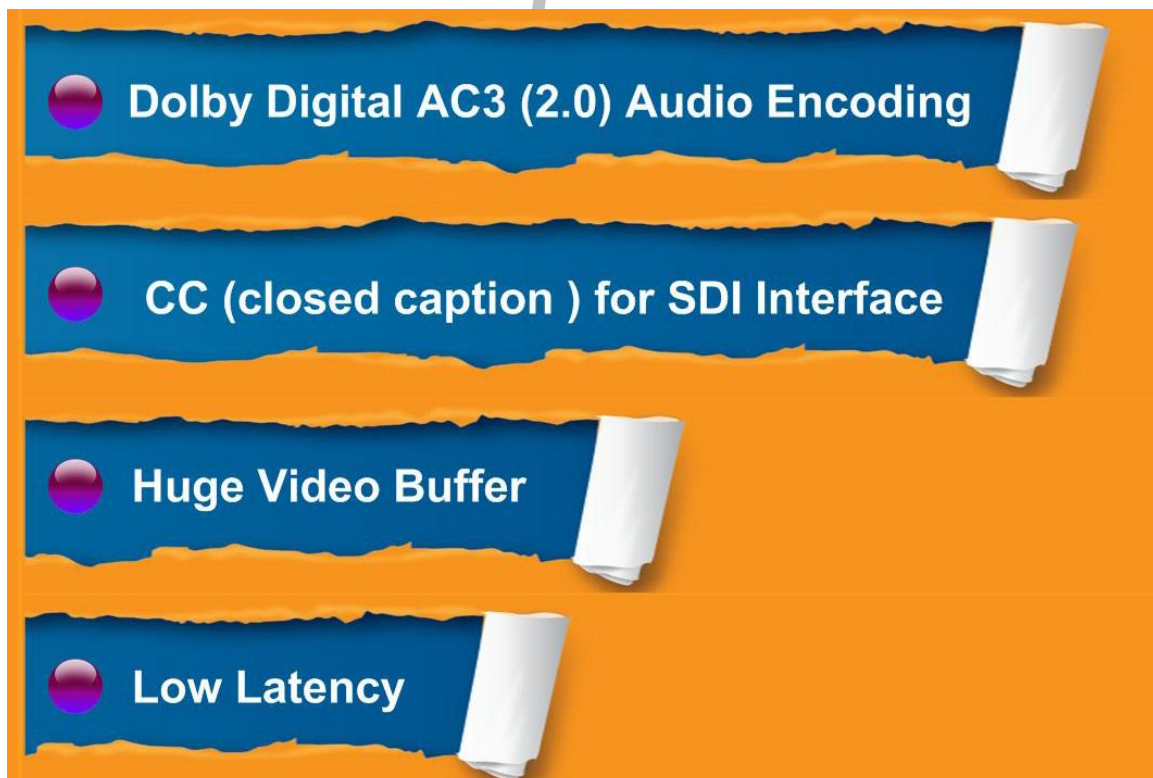


**FMUSER FUQ-4653B Encoder Modulator 4x HDMI-SDI Inputs
MPEG-2-4 AVC Encoding DVB-ATSC-ISDB-T Outputs IP-RF
Streaming Broadcast Headend IPTV**



Product overview

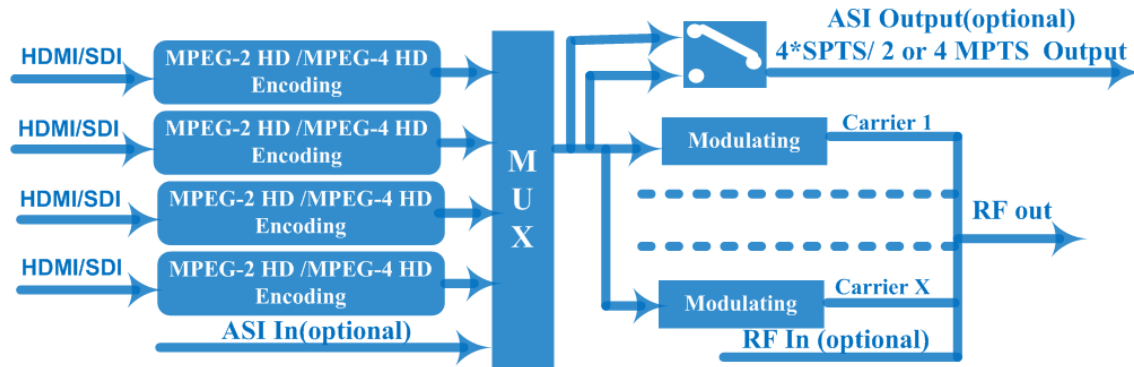
The FMUSER FUQ-4653B combines MPEG-2/MPEG-4 AVC encoding and DVB/ATSC/ISDB-T modulation in a 1RU chassis, accepting 4x HDMI/SDI and optional ASI inputs, while delivering IP/UDP/RTP streams or RF output (47-862MHz). It processes signals with <85ms latency in selectable modes and supports MPTS/SPTS

IP multiplexing. Compatible with HMS/SMS management systems, this hot-swappable unit is ideal for digital terrestrial TV headends and multi-screen IPTV distribution.

Key feature

Modular design: Hot-swappable encoder/ASI boards with tool-free maintenance

Principle Chart



Specifications

HDMI Encoder Board	Video	Input	HDMI*2/board	HDMI*2/board	HDMI*2/board
HDMI Encoder Board	Video	Encoding	MPEG2; MPEG4 AVC/H.264	MPEG2; MPEG4 AVC/H.264	MPEG2; MPEG4 AVC/H.264
HDMI Encoder Board	Video	Bitrate	1-19.5Mbps	1-19.5Mbps	1-19.5Mbps
HDMI Encoder Board	Video	Resolution	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i

HDMI Encoder Board	Video	Support object resolution (for downscale conversion)	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1440*1080_60i, 1440*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1440*1080_60i, 1440*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1440*1080_60i, 1440*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i
HDMI Encoder Board	Video	Low Delay	Normal, Mode 1, Mode 2, Manual	Normal, Mode 1, Mode 2, Manual	Normal, Mode 1, Mode 2, Manual
HDMI Encoder Board	Video	Chroma	4:2:0	4:2:0	4:2:0
HDMI Encoder Board	Video	Aspect Ratio	16:9,4:3	16:9,4:3	16:9,4:3
HDMI Encoder Board	Audio	Encoding	MPEG1 Layer II;MPEG 2-AAC; MPEG 4-AAC and Dolby Digital AC3 2.0(optional)	MPEG1 Layer II;MPEG 2-AAC; MPEG 4-AAC and Dolby Digital AC3 2.0(optional)	MPEG1 Layer II;MPEG 2-AAC; MPEG 4-AAC and Dolby Digital AC3 2.0(optional)
HDMI Encoder Board	Audio	Dialog Normalization Applicable for DD AC3 encoding only)	-31 ~ -1 dB	-31 ~ -1 dB	-31 ~ -1 dB
HDMI Encoder Board	Audio	Sample rate	48KHz	48KHz	48KHz
HDMI Encoder Board	Audio	Bitrate	64/96/128/ 192/256/320kbps	64/96/128/ 192/256/320kbps	64/96/128/ 192/256/320kbps

SDI Encoder Board					
Modulator Section	DVB-T (Optional)	Standard	DVB-T	DVB-T	DVB-T
Modulator Section	DVB-T (Optional)	Bandwidth	6M, 7M, 8M	6M, 7M, 8M	6M, 7M, 8M
Modulator Section	DVB-T (Optional)	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulator Section	DVB-T (Optional)	Code rate	1/2, 2/3, 3/4, 5/6, 7/8.	1/2, 2/3, 3/4, 5/6, 7/8.	1/2, 2/3, 3/4, 5/6, 7/8.
Modulator Section	DVB-T (Optional)	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulator Section	DVB-T (Optional)	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K	2K, 4K, 8K
Modulator Section	DVB-T (Optional)	MER	≥40dB	≥40dB	≥40dB
Modulator Section	DVB-T (Optional)	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulator Section	DVB-T (Optional)	RF Out	4*DVB-T carriers(40MHz bandwidth)	4*DVB-T carriers(40MHz bandwidth)	4*DVB-T carriers(40MHz bandwidth)
Modulator Section	DVB-T (Optional)	RF output level	-20~ -3dbm, 0.1db step	-20~ -3dbm, 0.1db step	-20~ -3dbm, 0.1db step
Modulator Section	DVB-C (Optional)	Standard	J.83A (DVB-C), J.83B, J.83C	J.83A (DVB-C), J.83B, J.83C	J.83A (DVB-C), J.83B, J.83C
Modulator Section	DVB-C (Optional)	MER	≥40dB	≥40dB	≥40dB
Modulator Section	DVB-C (Optional)	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulator Section	DVB-C (Optional)	RF output level	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step
Modulator Section	DVB-C (Optional)	Symbol rate	3.000~9.000 Msps adjustable	3.000~9.000 Msps adjustable	3.000~9.000 Msps adjustable
Modulator Section	DVB-C (Optional)	RF Out	4*DVB-C carriers(40MHz bandwidth)	4*DVB-C carriers(40MHz bandwidth)	4*DVB-C carriers(40MHz bandwidth)

Modulator Section	DVB-C (Optional)		J.83A	J.83B	J.83C
Modulator Section	DVB-C (Optional)	Constellation	16/32/64/128/256QAM	64/ 256 QAM	64/ 256 QAM
Modulator Section	DVB-C (Optional)	Bandwidth	8M	6M	6M
Modulator Section	ATSC (Optional) (under developing)	Standard	ATSC A/53	ATSC A/53	ATSC A/53
Modulator Section	ATSC (Optional) (under developing)	MER	≥40dB	≥40dB	≥40dB
Modulator Section	ATSC (Optional) (under developing)	RF frequency	50~960MHz, 1KHz step.	50~960MHz, 1KHz step.	50~960MHz, 1KHz step.
Modulator Section	ATSC (Optional) (under developing)	RF Out	4*ATSC carriers	4*ATSC carriers	4*ATSC carriers
Modulator Section	ATSC (Optional) (under developing)	RF output level	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step	-20~ +3dbm, 0.1db step
Modulator Section	ATSC (Optional) (under developing)	Constellation	8VSB	8VSB	8VSB
Modulator Section	ISDB-T (Optional)	Standard	ARIB STD-B31	ARIB STD-B31	ARIB STD-B31
Modulator Section	ISDB-T (Optional)	Bandwidth	6M	6M	6M
Modulator Section	ISDB-T (Optional)	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulator Section	ISDB-T (Optional)	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4

Modulator Section	ISDB-T (Optional)	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K	2K, 4K, 8K
Modulator Section	ISDB-T (Optional)	MER	≥40dB	≥40dB	≥40dB
Modulator Section	ISDB-T (Optional)	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulator Section	ISDB-T (Optional)	RF Out	2*ISDBT carriers(30MHz bandwidth)	2*ISDBT carriers(30MHz bandwidth)	2*ISDBT carriers(30MHz bandwidth)
Modulator Section	ISDB-T (Optional)	RF output level	-20~ -3dbm, 0.1db step	-20~ -3dbm, 0.1db step	-20~ -3dbm, 0.1db step
General	System	Local interface	LCD + control buttons	LCD + control buttons	LCD + control buttons
General	System	Remote management	Web NMS	Web NMS	Web NMS
General	System	Stream Out	2 ASI out (BNC type, same one mirror out ts as one of 4/2 MPTS outputs and 4 SPTS outputs)	2 ASI out (BNC type, same one mirror out ts as one of 4/2 MPTS outputs and 4 SPTS outputs)	2 ASI out (BNC type, same one mirror out ts as one of 4/2 MPTS outputs and 4 SPTS outputs)
General	System	Stream Out	DVB-C/DVB-T/ATSC: IP (4 MPTS & 4 SPTS) out over UDP, RTP/RTSP ISDBT: IP (2 MPTS & 4 SPTS) out over UDP, RTP/RTSP	DVB-C/DVB-T/ATSC: IP (4 MPTS & 4 SPTS) out over UDP, RTP/RTSP ISDBT: IP (2 MPTS & 4 SPTS) out over UDP, RTP/RTSP	DVB-C/DVB-T/ATSC: IP (4 MPTS & 4 SPTS) out over UDP, RTP/RTSP ISDBT: IP (2 MPTS & 4 SPTS) out over UDP, RTP/RTSP
General	System	DATA Port	1000M	1000M	1000M
General	System	NMS interface	RJ45, 100M	RJ45, 100M	RJ45, 100M
General	System	Language	English	English	English
General	Physical Specification	Power supply	100~240VAC, 50/60Hz	100~240VAC, 50/60Hz	100~240VAC, 50/60Hz

General	Physical Specification	Dimensions	482*400*44m m	482*400*44m m	482*400*44m m
General	Physical Specification	Operation temperature	0~45■	0~45■	0~45■

Major Technical Comparison

	The Former Version	The Current Version■V2■
Bit rate Mode	CBR/VBR option	CBR
Audio Group/Pair option-SDI	No	Yes
Low Delay	Normal/Mode 1/Mode 2	Normal/Mode 1/Mode 2/Manual
Character Encoding option	No	Yes
DATA Port	100M port	1000M port
Frequency Range	30-960MHz	50-960MHz
Symbol Rate--DVB-C	5-9Msps	3-9Msps
ISDB-T carriers	1 or 2 carriers for order option	2 carriers
IP out--DVB-C	1 MPTS and 4 SPTS	4 MPTS and 4 SPTS
IP out--DVB-T	4 SPTS	4 MPTS and 4 SPTS
DVB-T carriers	2 carriers	4 carriers

The other differences are existing Web GUI design and chassis design etc(NMS/DATA port position), no further description here.

Internal Test Report of Latency

The values cover the progress from Encoding → Decoding

Decoding Terminal	Encoding Details	Encoding Details	Encoding Details	Encoding Details	Encoding Details	Average Latency (ms)
Decoding Terminal	Resolution	Encoding Bit Rate	Low Latency Mode	Signal Source	Encoding Type	Average Latency (ms)
DVB-C HD STB	1080i@50	14M	MODE 1	HDMI	mpeg2	170
DVB-C HD STB	1080i@50	14M	MODE 1	HDMI	H.264	347.5

DVB-C HD STB	1080i@50	14M	MODE 1	SDI	mpeg2	227.5
DVB-C HD STB	1080i@50	14M	MODE 1	SDI	H.264	367.5
DVB-C HD STB	1080i@50	14M	MODE 2	HDMI	mpeg2	222.5
DVB-C HD STB	1080i@50	14M	MODE 2	HDMI	H.264	395
DVB-C HD STB	1080i@50	14M	MODE 2	SDI	mpeg2	240
DVB-C HD STB	1080i@50	14M	MODE 2	SDI	H.264	397.5
DVB-C HD STB	720p@50	14M	MODE 1	HDMI	mpeg2	85.75
DVB-C HD STB	720p@50	14M	MODE 1	HDMI	H.264	237.5
DVB-C HD STB	720p@50	14M	MODE 1	SDI	mpeg2	127.5
DVB-C HD STB	720p@50	14M	MODE 1	SDI	H.264	295
DVB-C HD STB	720p@50	14M	MODE 2	HDMI	mpeg2	182.5
DVB-C HD STB	720p@50	14M	MODE 2	HDMI	H.264	277.5
DVB-C HD STB	720p@50	14M	MODE 2	SDI	mpeg2	167.5
DVB-C HD STB	720p@50	14M	MODE 2	SDI	H.264	325
DVB-C HD STB	576i@50	14M	MODE 1	HDMI	mpeg2	310
DVB-C HD STB	576i@50	14M	MODE 1	HDMI	H.264	600
DVB-C HD STB	576i@50	14M	MODE 1	SDI	mpeg2	330
DVB-C HD STB	576i@50	14M	MODE 1	SDI	H.264	620
DVB-C HD STB	576i@50	14M	MODE 2	HDMI	mpeg2	270

DVB-C HD STB	576i@50	14M	MODE 2	HDMI	H.264	610
DVB-C HD STB	576i@50	14M	MODE 2	SDI	mpeg2	280
DVB-C HD STB	576i@50	14M	MODE 2	SDI	H.264	620

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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